

Proof. **Step1. Degenerate cases.** If $a = 0$ then $ab = 0$ and the right-hand side equals $\frac{b^q}{q} \geq 0$; thus the inequality holds, with equality only when also $b = 0$. The same argument applies when $b = 0$. Hence the statement is true whenever at least one of the numbers is zero, and equality occurs exactly in the case $a = b = 0$.

Step2. Convex function and its Legendre-conjugate. Define the function

$$\phi : [0, \infty) \rightarrow \mathbb{R}, \quad \phi(x) = \frac{x^p}{p},$$

with $p > 1$. Its first and second derivatives are

$$\phi'(x) = x^{p-1}, \quad \phi''(x) = (p-1)x^{p-2} > 0 \quad (x > 0),$$

so ϕ is strictly convex on $[0, \infty)$.

The convex (Legendre) conjugate of ϕ is

$$\phi^*(y) = \sup_{x \geq 0} \{xy - \phi(x)\}, \quad y \geq 0.$$

To find the supremum we differentiate the inner expression:

$$\frac{d}{dx}(xy - \phi(x)) = y - \phi'(x) = y - x^{p-1}.$$

The unique stationary point satisfies $x_0 = y^{1/(p-1)}$. Because ϕ is strictly convex, this point gives the global maximum. Substituting x_0 we obtain

$$\begin{aligned} \phi^*(y) &= y y^{1/(p-1)} - \frac{(y^{1/(p-1)})^p}{p} \\ &= y^{p/(p-1)} - \frac{y^{p/(p-1)}}{p} \\ &= \frac{p-1}{p} y^{p/(p-1)}. \end{aligned}$$

Define

$$q := \frac{p}{p-1} \quad \Longleftrightarrow \quad \frac{1}{q} = \frac{p-1}{p},$$

so that $\frac{1}{p} + \frac{1}{q} = 1$. Consequently

$$\boxed{\phi^*(y) = \frac{y^q}{q}} \quad (y \geq 0).$$

Step3. Application of the Fenchel-Young inequality. For any convex function ϕ and its conjugate ϕ^* , the Fenchel-Young inequality states

$$ab \leq \phi(a) + \phi^*(b), \quad a, b \geq 0.$$

Proof of Fenchel-Young. By definition of $\phi^*(b)$,

$$\phi^*(b) = \sup_{x \geq 0} (bx - \phi(x)) \geq ba - \phi(a).$$

Rearranging yields $ab \leq \phi(a) + \phi^*(b)$, completing the lemma.

Applying the lemma with the concrete ϕ and ϕ^* computed in Step2 gives

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q},$$

which is precisely Young's inequality.

Step4. Equality condition. In the Fenchel-Young inequality equality holds iff the supremum defining $\phi^*(b)$ is attained at $x = a$. The maximiser of $bx - \phi(x)$ is $x = b^{1/(p-1)}$; therefore equality requires

$$a = b^{1/(p-1)} \iff b = a^{p-1}.$$

Raising both sides to the power p yields the equivalent condition $a^p = b^q$. Together with the degenerate case handled in Step1, we obtain that equality occurs exactly when $(a, b) = (0, 0)$ or $a^p = b^q$ (equivalently $b = a^{p-1}$).

Step5. Elementary proof via weighted AM–GM (optional). The weighted arithmetic–geometric mean inequality states that for non-negative X, Y and weights $\alpha, \beta > 0$ with $\alpha + \beta = 1$,

$$X^\alpha Y^\beta \leq \alpha X + \beta Y.$$

Take $X = a^p$, $Y = b^q$, $\alpha = \frac{1}{p}$ and $\beta = \frac{1}{q}$; note that $\alpha + \beta = 1$ because $\frac{1}{p} + \frac{1}{q} = 1$. Then

$$(a^p)^{1/p} (b^q)^{1/q} \leq \frac{1}{p} a^p + \frac{1}{q} b^q,$$

i.e. $ab \leq \frac{a^p}{p} + \frac{b^q}{q}$. Equality in the weighted AM–GM occurs exactly when $a^p = b^q$, matching the condition found above.

Step6. Conclusion. For all non-negative real numbers a, b and exponents $p, q > 1$ satisfying $\frac{1}{p} + \frac{1}{q} = 1$,

$$\boxed{ab \leq \frac{a^p}{p} + \frac{b^q}{q}}.$$

The inequality is strict unless either $a = b = 0$ or $a^p = b^q$ (equivalently $b = a^{p-1}$). This completes the proof. $\square$ $\square$